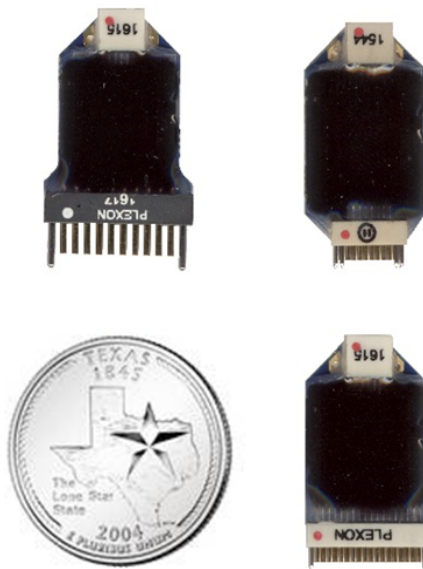


Digital Headstage Technical Guide

8, 16, 32, 64, and 128 Channel Digital Headstages



Digital Headstage Technical Guide

Contents

4	Introduction
5	Digital Headstage Cables
7	Digital Headstage L - 8 channel, 0.050" pitch
9	Digital Headstage L - 16 channel, 0.050" pitch
11	Digital Headstage Q – 16 channel, 0.025" pitch (Legacy Item)
13	Digital Headstage Q2 – 16 channel, 0.025" pitch Gen2
15	Digital Headstage R – 32 channel, 0.025" pitch Gen2
17	Digital Headstage S – 64 channel, 0.025" pitch
19	Digital Headstage T – 16 channel, 0.025" pitch Gen2
21	Digital Stimulating Headstage ST - 12 channel, 0.025" pitch
23	Digital Headstage U – 64 channel (Samtec connector, thumbtack design)
25	Digital Headstage V - 128 channel Gen 1 (Samtec connector, thumbtack design)
26	Digital Headstage V2 - 128 channel Gen 2 (Samtec connector, thumbtack design)
28	Headstage and Conversion Options
28	Light Emitting Diode (LED)
28	Latching
29	Digital Headstage Processor Connector Pin-out, Ground and Reference Information
29	Matching Digital Headstage Models to OmniPlex Settings

Documentation History

Date	Version	Notes
March 2021	DHSTTN0001h	- Added HST/12-NLT
August 2020	DHSTTN0001g	- Added HST/128DS (Samtec connector, thumbtack design)
September 2019	DHSTTN0001f	- Small edits to input connectors
July 2019	DHSTTN0001e	- Added HST/64DS (Samtec connector, thumbtack design)
September 2018	DHSTTN0001d	- Update pinouts
December 2017	DHSTTN0001c	- Update pinouts
January 2017	DHSTTN0001b	- Update pinouts
October 2016	DHSTTN0001a	- First issue of the Digital Headstage Technical Guide.

Introduction

This Digital Headstage Technical Guide describes the digital headstages that are available from Plexon®, along with connector pin-out configurations and ground and reference information. These headstages are designed for use with the Plexon Digital Headstage Processor (DHP) unit, sold separately.

DIGITAL HEADSTAGE CABLE:

Special cables are used to connect the digital headstages to the Plexon DHP unit. Plexon 8, 16, and 32 channel digital headstages connect through the Plexon HSC/DHSC1 or HSC/DHSC3 cable. Plexon 64 and 128 channel headstages connect through the Plexon HSC/DHSC2 or HSC/DHSC4 cable. These cables are described in this document.

REFERENCE SIGNAL:

Each headstage has at least one Reference pin that can be used to acquire a reference signal from a de-insulated low-impedance electrode. The reference signal can be used to subtract common mode noise and artifacts from the signals on the neural recording channels. For all digital headstages except the 64 and 128 channel headstage, the Reference pin may be internally grounded by selecting the “Grounded reference” setting in the Plexon OmniPlex® Neural Data Acquisition System (sold separately).

Some connectors have multiple Reference pins. On any headstage, you only need to connect one reference electrode and you may connect it to any one of the Reference pins. If you have no reference electrode connected, then you should operate the headstage in “Grounded reference” mode.

OPTIONS:

These options can be added to the headstages:

- Mounted LEDs for use in behavioral analysis research using Plexon’s CinePlex® and CineLyzer® Behavioral Research Systems. The LEDs are available in red, blue, and green (Available on all digital headstages).
- Latching mechanism (clip) to enhance headstage retention with more active animals (Available on headstages with 0.050 pitch input connectors).

NOTES:

The product images in this document are not to scale relative to either actual size or to each other.

The headstage lengths include the longest pin.

The headstage weights may vary slightly due to the hand crafted nature of the products.

The addition of LEDs and/or a latch will impact the weight or increase the dimensions of the headstage slightly.

Digital Headstage Cables

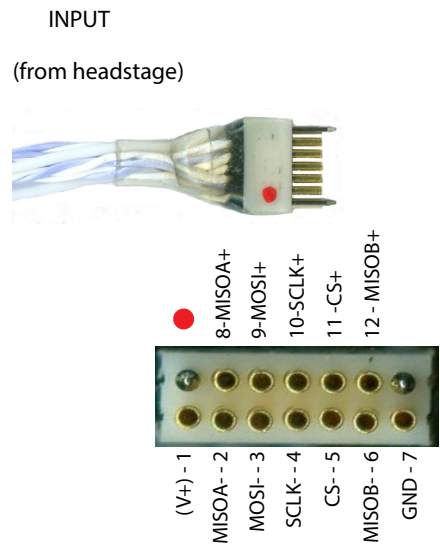
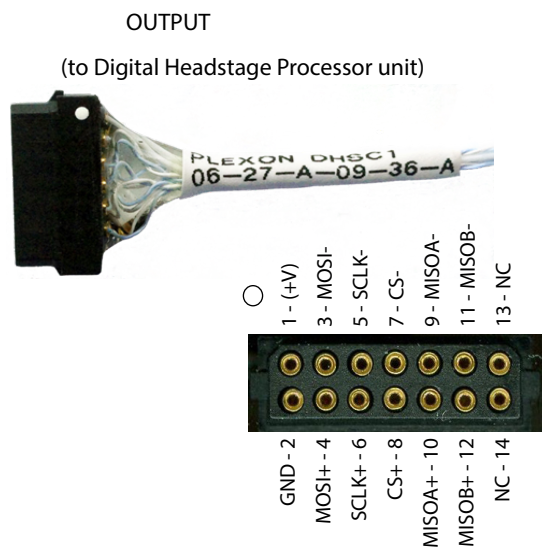
Catalog Number: HSC/DHSC1:

A cable designed for use with any of the Plexon 8, 16, and 32 channel digital headstages.

Lengths

Lengths up to 243cm (96in) are available. The default length is 91cm (36in).

Headstage Cable Views



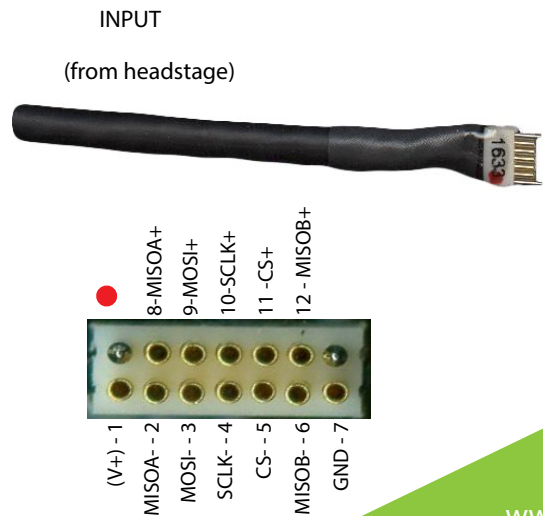
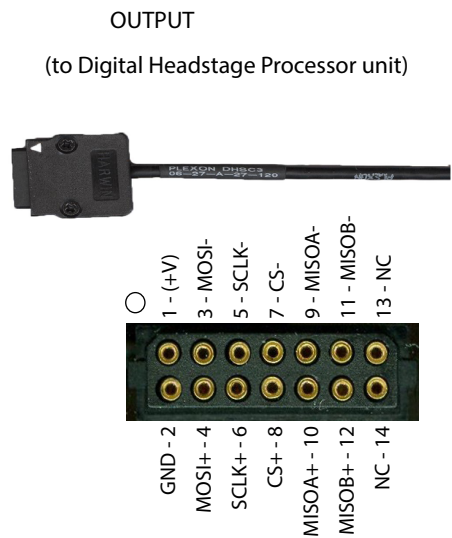
Catalog Number: HSC/DHSC3

A cable designed for use with any of the Plexon 8, 16, and 32 channel digital headstages.

Lengths

Lengths up to 305cm (120in) are available. The default length is 91cm (36in).

Headstage Cable Views



Digital Headstage Cables

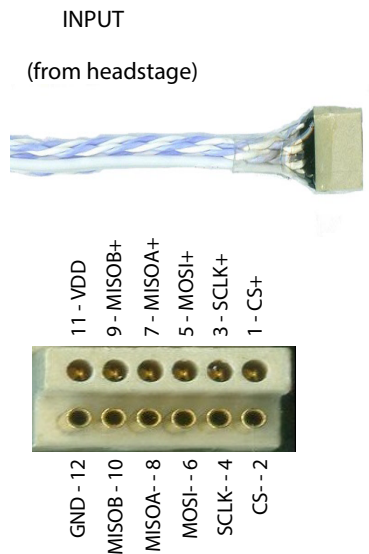
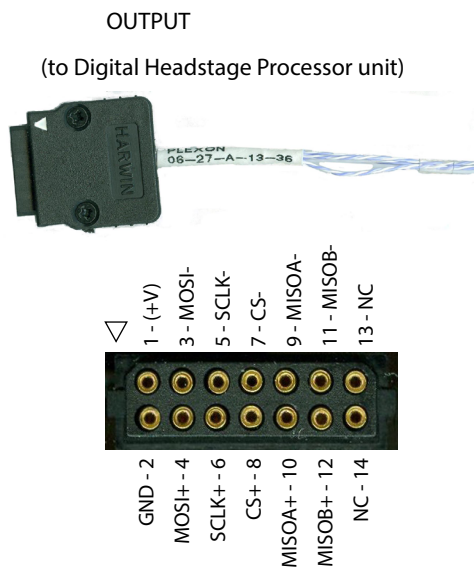
Catalog Number: HSC/DHSC2

A cable designed for use with the Plexon 64 and 128 channel digital headstages.

Lengths

Lengths up to 243cm (96in) are available. The default length is 91cm (36in).

Headstage Cable Views

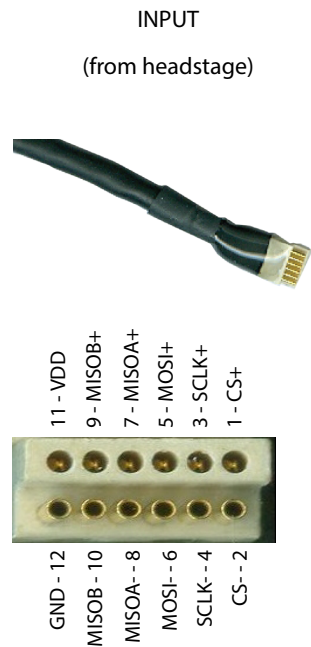
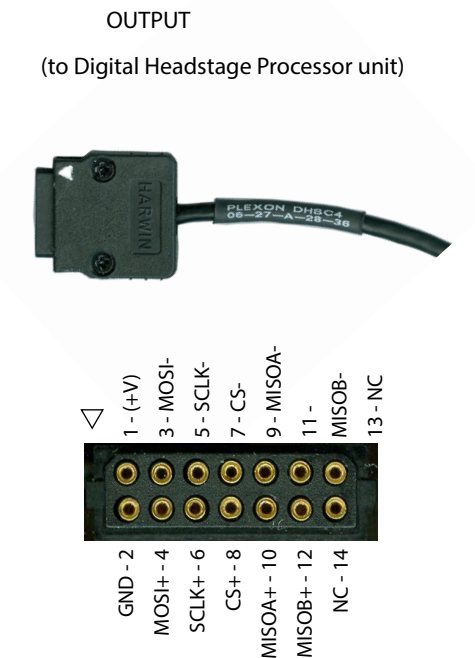


Catalog Number: HSC/DHSC4

A cable designed for use with the Plexon 64 and 128 channel digital headstages.

Lengths

Lengths up to 243cm (96in) are available. The default length is 91cm (36in).



Headstage L – 8 channel, 0.050" pitch Digital Headstage



Channels:	8	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Pitch:	0.050" (1.27mm)	Dimensions:	1.08L x 0.62W x 0.09D inches (27.4L x 15.8W x 2.3D mm)
Reference pins:	1	Weight:	0.04oz (1.2g)
Grounding pins:	1		
Guide pins:	2		
Catalog Number:	HST/8o50D Gen2		
Reference:	Dedicated reference; programmably groundable		
Power Supply:	Powered from Digital Headstage Processor (DHP)		
Input Impedance:	13M Ω @ 1kHz		

Description:

An 8 channel digital headstage amplifier containing eight channel input pins, one reference pin, and one grounding pin. The distance between pins (the pitch) is 0.050" (1.27mm).

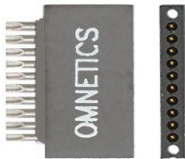
Input Connector Information: Connector #1

Input Mating Connector:

Male

Catalog Number:

CON/8o50m-10P

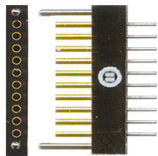


Input Connector:

Female

Catalog Number:

CON/8o50f-10P



Headstage Cable Information:

All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 or HSC/DHSC3 cable.

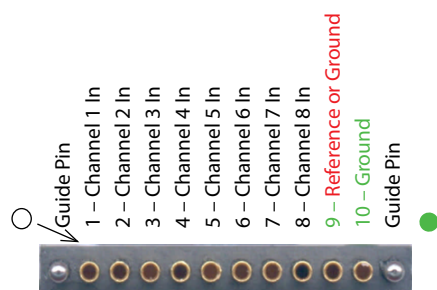
Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

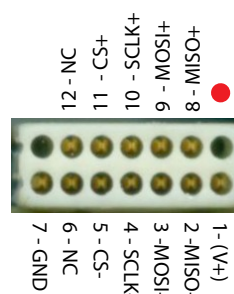
Channels 1-8 are designed for recording from electrodes. Pin 9 is used as a reference input. This reference signal can be used to subtract common mode noise and artifacts from the 8 recording channels. The processor in the headstage subtracts the reference signal (on Pin 9) from the signals on each of the 8 channels, then outputs the resulting signal.

Catalog Number: HST/8o50D Gen2

INPUT



OUTPUT



Input Pin 9: Reference or Ground

- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to “True reference” to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to “Grounded reference” to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.

Input Pin 10: Ground

- ◆ Use this pin to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage L – 16 channel, 0.050" pitch Digital Headstage



Channels:	16	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Pitch:	0.050" (1.27mm)		
Reference pins:	2	Dimensions:	1.08L x 0.62W x 0.15D inches (27.4L x 15.8W x 3.8D mm)
Grounding pins:	2		
Guide pins:	4		
Catalog Number:	HST/16o50D Gen2	Weight:	0.06oz (1.7g)
Reference:	Dedicated reference; programmably groundable		
Power Supply:	Powered from Digital Headstage Processor (DHP)		
Input Impedance:	13M Ω @ 1kHz		

Description:

A 16 channel digital headstage amplifier containing 16 channel input pins, two reference pins, and two grounding pins.

tNote: This headstage uses two of these 8-channel connectors, stacked and glued together.

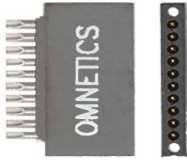
Input Connector Information: Connector #1

Input Mating Connector:

Male

Catalog Number:

CON/8o50m-10P

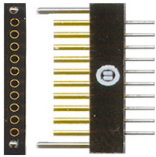


Input Connector:

Female

Catalog Number:

CON/8o50f-10P



Headstage Cable Information:

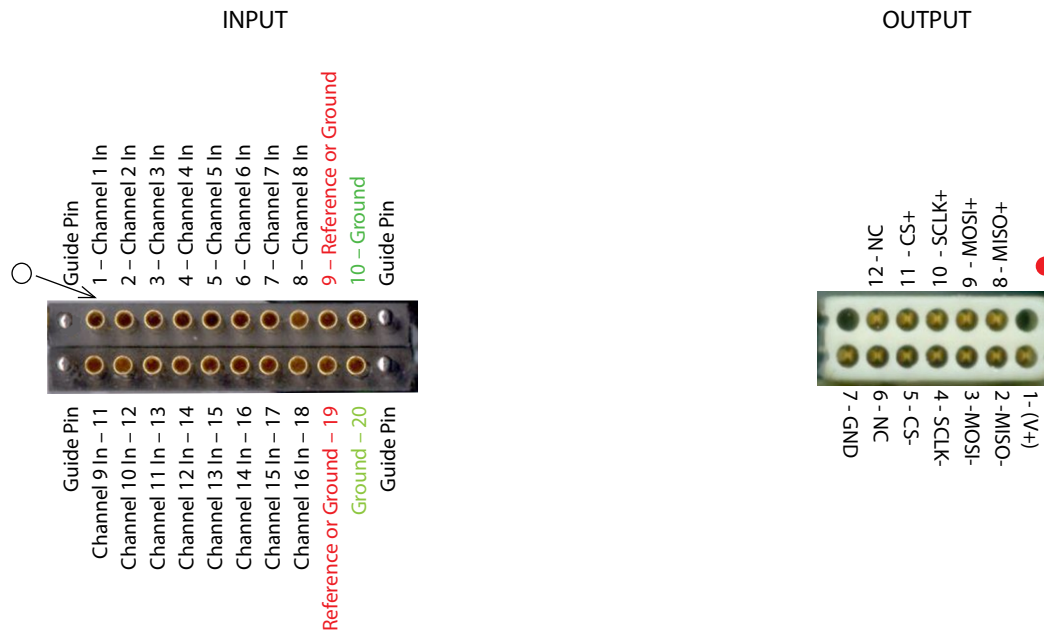
All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 or HSC/DHSC3 cable.

Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

Channels 1-16 are designed for recording from electrodes. Pins 9 and 19 can be used as a reference input. This reference signal can be used to subtract common mode noise and artifacts from the 16 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 16 channels, then outputs the resulting signal. See the additional discussion about Pins 9 and 19, below.

Catalog Number: HST/16o50D Gen2



Input Pin 9 and 19: Reference or Ground

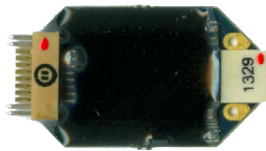
- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to "True reference" to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to "Grounded reference" to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.
- ◆ You only need to connect one reference electrode and you may connect it to either of the Reference pins. If you have no reference electrode connected, then you should operate the headstage in "Grounded reference" mode.

Input Pin 10 and 20: Ground

- ◆ The ground on Pins 10 and 20 are identical. Use either or both of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

HST/16D – 16 channel, 0.025” pitch Digital Headstage

(Legacy item, formerly designated Headstage Q)



Channels:	16	Amplifier/Processor Compatibility:
Pitch:	0.025" (.64mm)	Digital Headstage Processor (DHP)
Reference pins:	1	Dimensions:
Grounding pins:	1	0.97L x 0.52W x 0.10D inches
Guide pins:	6	(24.6L x 13.2W x 2.5D mm)
Catalog Number:	HST/16D	Weight:
Reference:	Dedicated reference; programmably groundable	0.035oz
Power Supply:	Powered from Digital Headstage Processor (DHP)	(1.0g)
Input Impedance:	13MΩ @ 1kHz	

Description:

This is a legacy part, not currently orderable, originally listed under catalog name HST/16D. It is a 16 channel digital headstage amplifier containing 16 channel input pins, one reference pin, and one grounding pin. The distance between pins (the pitch) is 0.025in.

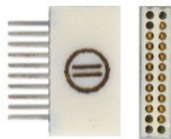
Input Connector Information: Connectors #6 and 7 (Nickel-free)

Input Mating Connector 6:

Male

Catalog Number:

CON/16m-V

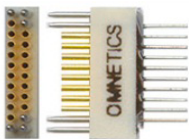


Input Connector 6:

Female

Catalog Number:

CON/16f-V

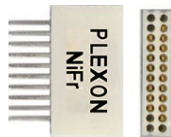


Input Mating Connector 7:

Male

Catalog Number:

CON/16m-V-NiFree
(MRI Compatible)



Headstage Cable Information:

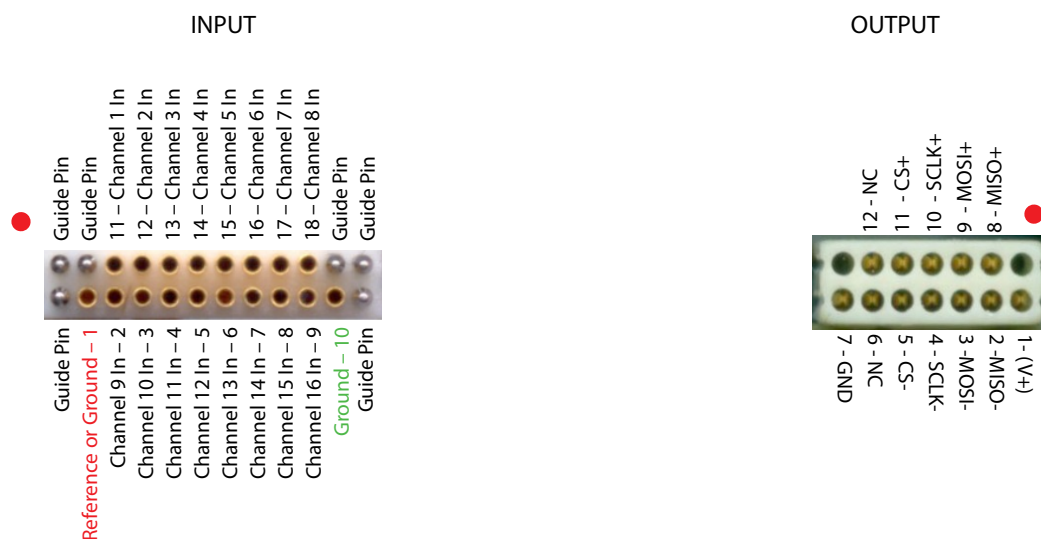
All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 or HSC/DHSC3 cable.

Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

Channels 1-16 are designed for recording from electrodes. Pin 1 can be used as a reference input. This reference signal can be used to subtract common mode noise and artifacts from the 16 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 16 channels, then outputs the resulting signal. See the additional discussion about Pins 1 and 10, below.

Catalog Number: HST/16D



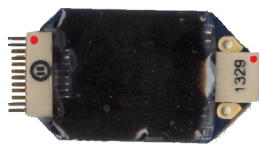
Input Pin 1 : Reference or Ground

- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to "True reference" to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to "Grounded reference" to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.

Input Pin 10: Ground

- ◆ Use this pin to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage Q2- HST/16D Gen2– 16 channel, 0.025” pitch Digital Headstage



Channels:	16	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Pitch:	0.025" (0.64 mm)	Dimensions:	1.00L x 0.51W x 0.10D inches (25.4L x 13.0W x 2.5D mm)
Reference pins:	1	Weight:	0.035oz (1.0g)
Grounding pins:	1		
Guide pins:	6		
Catalog Number:	HST/16D Gen2		
Reference:	Dedicated reference; programmably groundable		
Power Supply:	Powered from Digital Headstage Processor (DHP)		
Input Impedance:	13MΩ @ 1kHz		

Description:

A 16 channel digital headstage amplifier containing 16 channel input pins, one reference pin, and one grounding pin. The distance between pins (the pitch) is 0.025in.

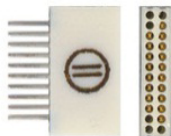
Input Connector Information: Connectors #6 and 7 (Nickel-free)

Input Mating Connector 6:

Male

Catalog Number:

CON/16m-V

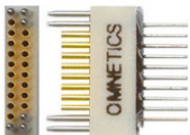


Input Connector 6:

Female

Catalog Number:

CON/16f-V

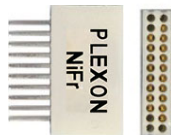


Input Mating Connector 7:

Male

Catalog Number:

CON/16m-V-NiFree
(MRI Compatible)



Headstage Cable Information:

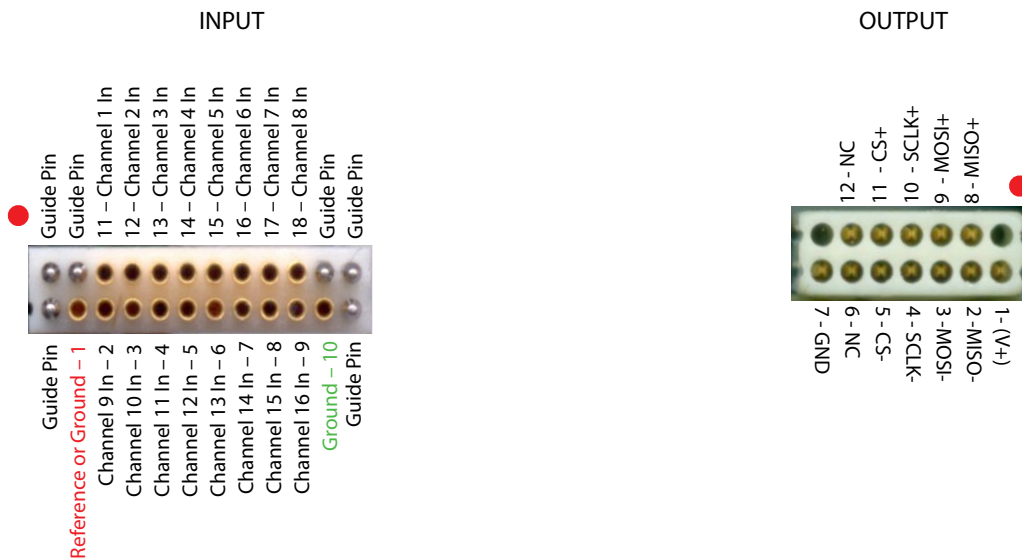
All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 of HSC/DHSC3 cable.

Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

Channels 1-16 are designed for recording from electrodes. Pin 1 can be used as a reference input. This reference signal can be used to subtract common mode noise and artifacts from the 16 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 16 channels, then outputs the resulting signal. See the additional discussion about Pins 1 and 10, below.

Catalog Number: HST/16D Gen2



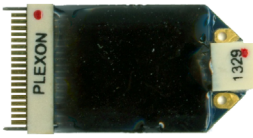
Input Pin 1: Reference or Ground

- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to "True reference" to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to "Grounded reference" to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.

Input Pin 10: Ground

- ◆ Use this pin to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage R – 32 channel, 0.025" pitch Digital Headstage



Channels:	32	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Pitch:	0.025" (.64mm)	Dimensions:	0.97L x 0.52W x 0.10D inches (24.6L x 13.2W x 2.5D mm)
Reference pins:	1	Weight:	0.039oz (1.1g)
Grounding pins:	3		
Guide pins:	4		
Catalog Number:	HST/32D		
Reference:	Dedicated reference; programmably groundable		
Power Supply:	Powered from Digital Headstage Processor (DHP)		
Input Impedance:	13MΩ @ 1kHz		

Description:

A 32 channel digital headstage amplifier containing 32 channel input pins, one reference pin, and three grounding pins. The distance between pins (the pitch) is 0.025in.

Input Connector Information: Connectors #8 and #9 (Nickel-free)

Input Mating Connector 8:

Male

Catalog Number:

CON/32m-V

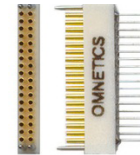


Input Connector 8:

Female

Catalog Number:

CON/32f-V



Input Mating Connector 9:

Male

Catalog Number:

CON/32m-V-NiFree
(MRI Compatible)



Headstage Cable Information:

All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 or HSC/DHSC3 cable.

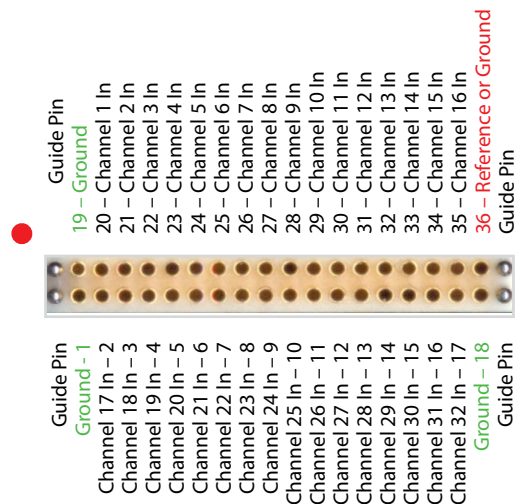
Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

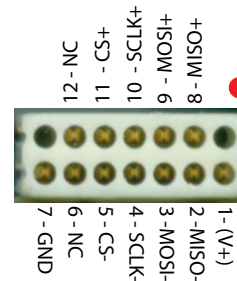
Channels 1-32 are designed for recording from electrodes. Pin 36 can be used as a reference input. This reference signal can be used to subtract common mode noise and artifacts from the 32 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 32 channels, then outputs the resulting signal. See the additional discussion about the reference and grounding pins, below.

Catalog Number: HST/32D

INPUT



OUTPUT



Input Pin 36: Reference or Ground

- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to "True reference" to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to "Grounded reference" to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.

Input Pin 1, 18 and 19: Ground

- ◆ The ground on Pins 1, 18 and 19 are identical. Use any or all of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage S – 64 channel, 0.025" pitch Digital Headstage



Channels:	64	Amplifier/Processor Compatibility: Digital Headstage Processor (DHP)
Pitch:	0.025" (0.64mm)	
<u>Grounded Reference Version</u>		
Reference pins:	0	Dimensions: 1.02L x 0.55W x 0.18D inches (25.9L x 14.0W x 4.6D mm)
Grounding pins:	8	
<u>True Reference Version</u>		
Reference pins:	4	Weight: 0.051oz (1.8g)
Grounding pins:	4	
Guide pins:	8	
Catalog Number:	HST/64D	
Reference:	Hardware-based; Not selectable in software	
Power Supply:	Powered from Digital Headstage Processor (DHP)	
Input Impedance:	13MΩ @ 1kHz	

Description:

A 64 channel digital headstage amplifier containing 64 channel input pins.

This headstage is available in two versions – True reference (labelled “TR”) and Grounded reference (labelled “GR”). The True reference version has four reference pins and four grounding pins. The Grounded reference version has no reference pins and eight grounding pins.

Input Connector Information: Connectors #8 and #9 (Nickel-free)

Note: Use two of these 32-channel connectors. If you are building an assembly, observe the gap and center-to-center dimension as shown in the diagram on the next page.

Input Mating Connector 8:

Male

Catalog Number:

CON/32m-V

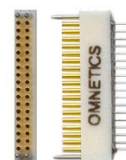


Input Connector 8:

Female

Catalog Number:

CON/32f-V

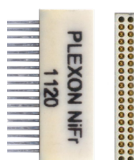


Input Mating Connector 9:

Male

Catalog Number:

CON/32m-V-NiFree
(MRI Compatible)



Headstage Cable Information:

The 64 channel digital headstage connects to the DHP unit through the HSC/DHSC2 cable.

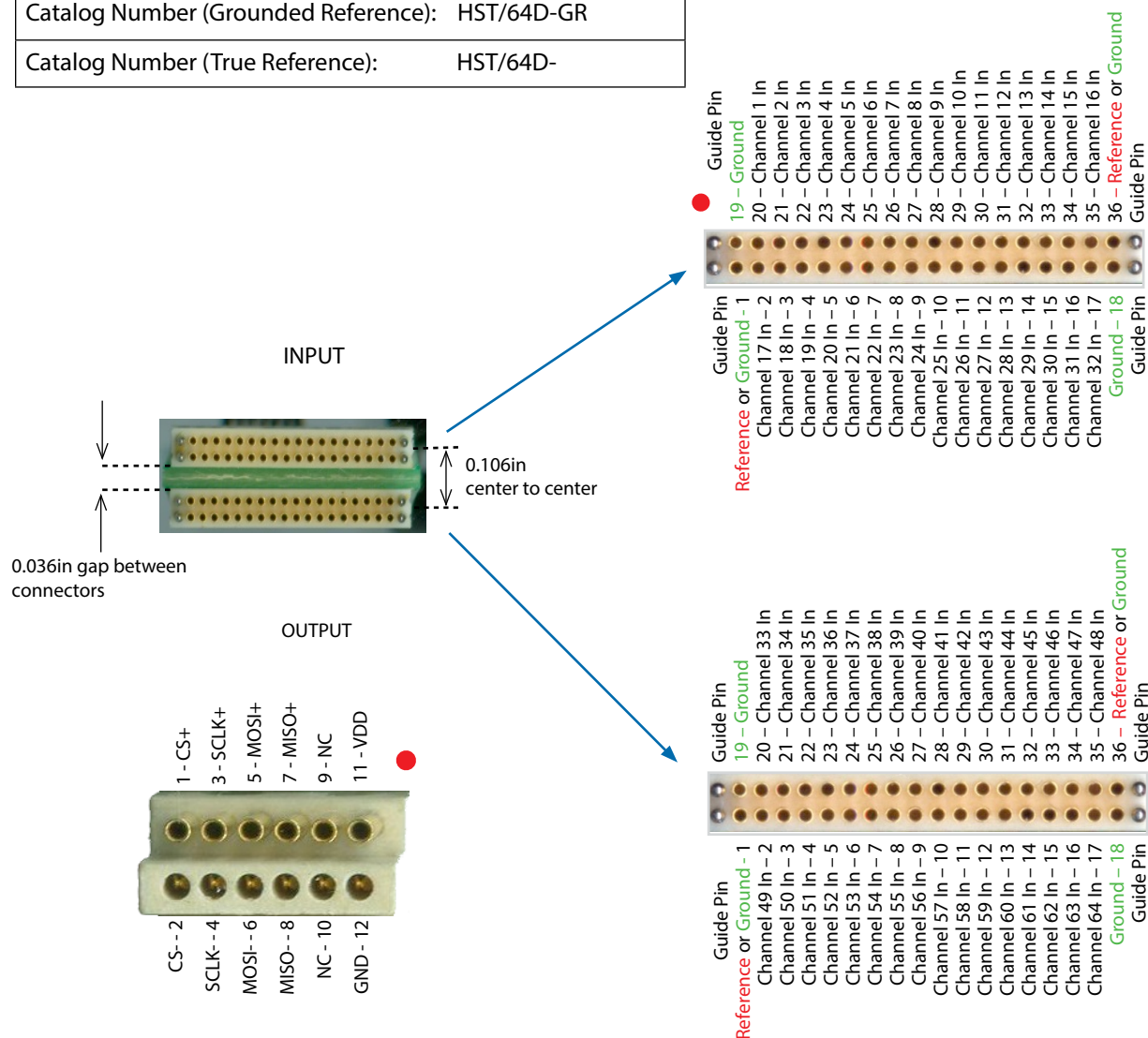
Connector Pin-out, Ground and Reference Information

User Selectable Reference

Channels 1-64 are designed for recording from electrodes. This headstage is available in two versions- True reference and Grounded reference. Pins 1 and 36 on both sides of the connector are pre-set in hardware as either Reference or Ground.

- ♦ In the True reference version, Pins 1 and 36 are set as Reference, so the connector has four reference pins and four grounding pins
- ♦ In the Grounded reference version, Pins 1 and 36 are set as Ground, so the connector has no reference pins and eight grounding pins.

Catalog Number (Grounded Reference):	HST/64D-GR
Catalog Number (True Reference):	HST/64D-



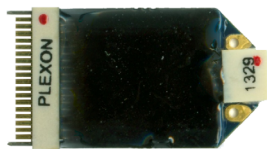
Input Pins 1 and 36 (2 places each): Reference or Ground

- ♦ See the INPUT diagram, above. In the True reference version, Pins 1 and 36 are electrically isolated from the Ground pins (Pins 18 and 19). In the Grounded reference version, Pins 1 and 36 are electrically shorted to the Ground pins (Pins 18 and 19).
- ♦ Pins 1 and 36 can be used to generate a reference signal from a de-insulated low-impedance electrode. This reference signal can be used to subtract common mode noise and artifacts from the 64 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 64 channels, then outputs the resulting signal
- ♦ You only need to connect one reference electrode and you may connect it to either of the Reference pins.

Input Pin 1, 18 and 19: Ground

- ♦ The ground on Pins 1 and 36 are identical. Use any or all of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage T- HST/16D Gen2 DIFF – 16 channel, 0.025” pitch fully differential reference Digital Headstage



Channels:	16	Amplifier/Processor Compatibility:
Pitch:	0.025" (0.64mm)	Digital Headstage Processor (DHP)
Reference pins:	16	Dimensions:
Grounding pins:	4	0.97L x 0.52W x 0.10D inches
Guide pins:	4	(24.6L x 13.2W x 2.5D mm)
Catalog Number:	HST/16D Gen2 DIFF	Weight:
Reference:	Dedicated	0.039oz
	Not selectable in software	(1.1g)
Power Supply:	Powered from Digital Headstage Processor (DHP)	
Input Impedance:	13MΩ @ 1kHz	

Description:

A 16 channel fully differential digital headstage amplifier containing 16 channel input pins, 16 dedicated reference pins, and four grounding pins. The distance between pins (the pitch) is 0.025in.

Input Connector Information: Connectors #8 and 9 (Nickel-free)

Input Mating Connector 8:

Male

Catalog Number:

CON/32m-V



Input Connector 8:

Female

Catalog Number:

CON/32f-V



Input Mating Connector 9:

Male

Catalog Number:

CON/32m-V-NiFree
(MRI Compatible)



Headstage Cable Information:

All Plexon 8, 16, and 32 channel digital headstages connect to the DHP unit through the HSC/DHSC1 or HSC/DHSC3 cable.

Connector Pin-out, Ground and Reference Information

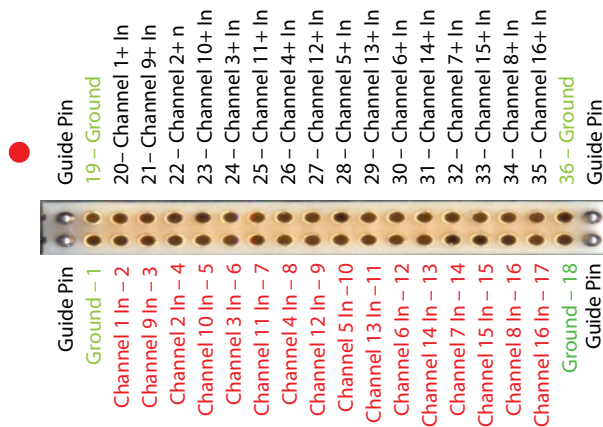
The input and output connectors for this headstage are shown in the image below. On the input connector:

- ♦ Pins 20-35 are designed for recording neural data from electrodes. The 16 channels are labeled with a "+" sign: Channel 1+ In, Channel 2+ In, and so forth. These channels are carrying the neural data.
- ♦ Pins 2-17 can be used as a reference input for each of the 16 channels. These 16 channels are labeled with a "-" sign: Channel 1- In, Channel 2- In, and so forth. Each reference channel is dedicated to the corresponding data channel. For example:
 - o Channel 1- In on Pin 2 is the reference signal that the system subtracts from Channel 1+ In on Pin 20
 - o Channel 16- In on Pin 17 is the reference signal that the system subtracts from Channel 16+ In on Pin 35

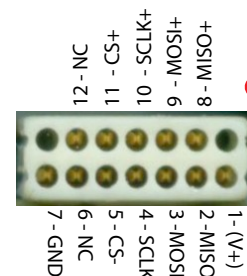
These dedicated reference signals can be used to subtract common mode noise and artifacts from the 16 recording channels. The processor in the headstage subtracts the reference signal from the signals on each of the 16 channels, then outputs the resulting signal.

Catalog Number: HST/16D Gen2 DIFF

INPUT



OUTPUT



Input Pin 2-17: Reference

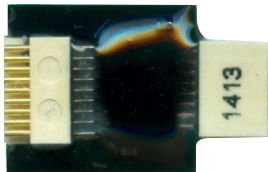
- ♦ These inputs can be used as channel specific reference electrodes. Each reference pin is dedicated to a specific channel pin.

Input Pins 1, 18, 19 and 36: Ground

- ♦ The ground on Pins 1, 18, 19, and 36 are identical. Use any or all of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Headstage ST- HST/12-NLT – 12 channel, 0.025" pitch

Digital Stimulating Headstage



Channels:	12	Amplifier/Processor Compatibility:	NeuroLight Stimulator
Pitch:	0.025" (0.64mm)	Dimensions:	0.67L x 0.39W x 0.18D inches (17L x 10W x 4.5D mm)
Reference pins:	N/A	Weight:	0.028oz (0.8g)
Stim return pins:	4	Catalog Number:	HST/12-NLT
Guide pins:	0	Reference:	N/A
		Power Supply:	Powered from NeuroLight Stimulator

Description:

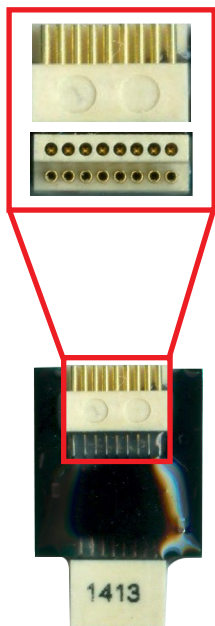
A 12 channel digital headstage containing 12 stimulation sources and four returns. The distance between pins (the pitch) is 0.025in. The headstage output controls 12 micro-LEDs independently.

Connector Information: Connector #14

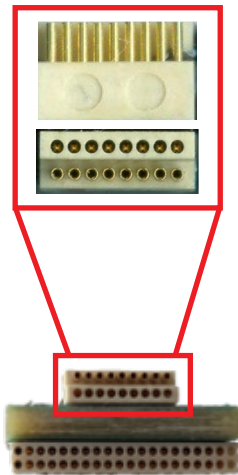
Catalog Number:
CON/18-9M-9F

The same connector is used for the optrode connector on the headset and the connector on the optrode.

Headstage HST/12-NLT



NeuroLight Optoelectrode



Headstage Cable Information:

Connects directly to the NeuroLight stimulator or to an R2S2 commutator through HSC/12-NLT.

Connector Pin-out

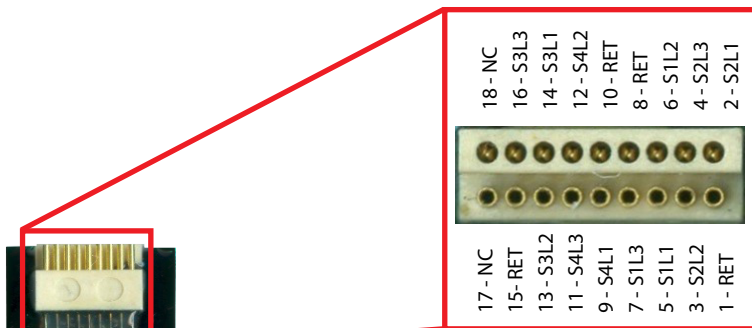
The connectors for this headstage are shown in the image below. On the headstage optoelectrode connector:

- ◆ Pins 1, 8, 10, and 15 are dedicated stim returns.

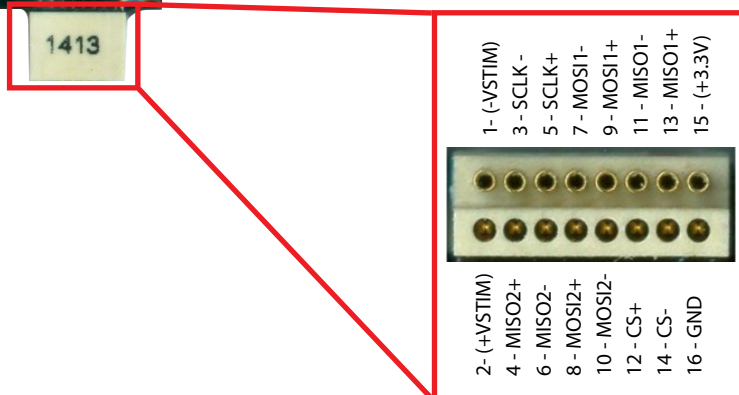
Pin #	LED Shank	LED Position
2	S2	L1
3	S2	L2
4	S2	L3
5	S1	L1
6	S1	L2
7	S1	L3
9	S4	L1
11	S4	L3
12	S4	L2
13	S3	L2
14	S3	L1
16	S3	L3

Catalog Number: HST/12-NLT

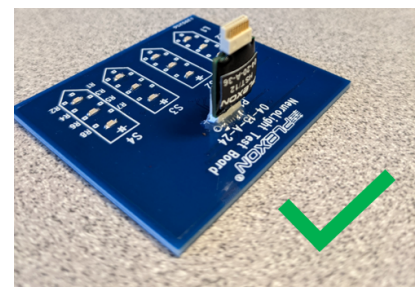
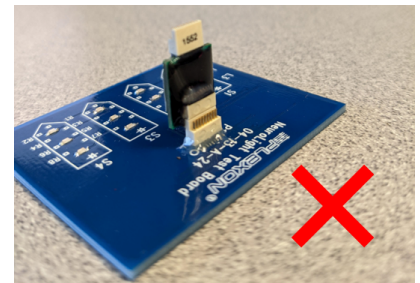
Headstage Optoelectrode Connector



Headstage Cable Connector



When plugging the headstage into the optoelectrode connector the pins should be aligned so that there are no visible gold pins after the connection is made.



Headstage U- HST/64DS – 64 channel Digital Headstage (Samtec connector, thumbtack design)



Channels:	64	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Connector:	Samtec (see below)		
Reference pins:	4	Dimensions:	0.47L x 0.55W x 0.61D inches (12L x 14W x 15.5D mm)
Grounding pins:	4	Weight:	0.051oz (1.45g)
Guide pins:	0		
Catalog Number:	HST/64DS		
Reference:	Dedicated reference; programmably groundable		
Power Supply:	Powered from Digital Headstage Processor (DHP)		
Input Impedance:	13MΩ @ 1kHz		

Description:

A 64-channel thumbtack design digital headstage with a Samtec connector interface. HST/64DS is also alternatively available with on board red, green, or blue LEDs (in any combination) or with LED sockets for the ability to add LEDs in the future. Custom electrode interface boards are available for HST/64DS.

Input Connector Information: Single piece Samtec connector #12

Note: The single piece Samtec connector requires no special alignment. The mating force with this connector is considerably less than the combined Omnectics connectors, without sacrificing a solid and reliable feel.

Input Mating Connector 12:

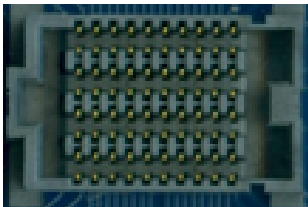
Female

Plexon Catalog Number:

CON/64f-Samtec

Samtec Part Number:

SEAF8-10-05.0-S-08-2-k



Input Connector 12:

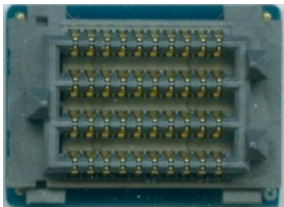
Male

Catalog Number:

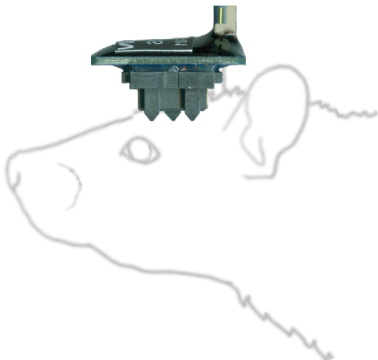
CON/64m-Samtec

Samtec Part Number:

SEAM8-10-S02.0-S-08-2-K



Thumbtack Design:



Headstage Cable Information:

The HST/64DS headstage connects to the DHP unit through the HSC/DHSC2 or HSC/DHSC4 cable.

Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

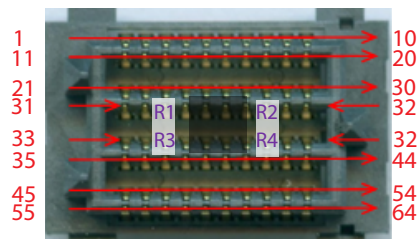
The figure on the left below shows which pins on the connector are recording inputs, which pins are connected to the (single) reference input, which pins are connected to ground, and which pins are not connected. The figure on the right below shows the Plexon channel numbers assigned to each of the recording inputs.

Catalog Number: HST/64DS

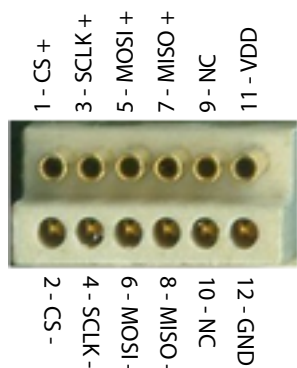
INPUT



- Recording Inputs
- Reference Input
- Ground
- No Connect



OUTPUT



Input Reference: Pins R1, R2, R3, and R4

- ◆ This input can be used to connect a dedicated reference electrode. It can be grounded programmatically from OmniPlex Server if no reference electrode is present or if you want to disregard the signal from the reference (for example, if the reference electrode is broken).
- ◆ Set the reference option in OmniPlex Server to "True reference" to subtract the signal from the reference electrode. It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- ◆ Set the reference option in OmniPlex Server to "Grounded reference" to ground the input when no reference electrode is present or when you want to disregard the signal from the reference electrode. This is typically used as a distant reference for field potentials, and also for spike channels.

Input Ground: Pins G1, G2, G3, and G4

- ◆ The ground on G1, G2, G3 and G4 are identical. Use any or all of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

Note

- ◆ When using a version of OmniPlex prior to 17.0, please contact Plexon Support for a channel mapping file

HST/128DS Gen 1 – 128 channel Digital Headstage (Samtec connector, thumbtack design) (Legacy item formerly designated Headstage V)



Channels:	128	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Connector:	Samtec (see below)	Dimensions:	0.83L x 0.47W x 0.39D inches (21L x 12W x 10D mm)
Reference pins:	16	Weight:	0.062oz (1.75g)
Grounding pins:	16	Power Supply:	Powered from Digital Headstage Processor (DHP)
Guide pins:	0	Input Impedance:	30M Ω @ 20kHz
Catalog Number:	HST/128DS Gen 1		
Reference:	Dedicated reference; programmably groundable		

Description:

A 128-channel thumbtack design digital headstage with a Samtec connector interface. HST/128DS Gen 1 is also alternatively available with on board red, green, or blue LEDs (in any combination) or with LED sockets for the ability to add LEDs in the future. Custom electrode interface boards are available for HST/128DS Gen 1.

Input Connector Information: Samtec connector #13a

Note: The input connector is a pair of Samtec ADM6-20-01.5-L-04-2 Gen 1 connectors. The connectors are spaced 6.35 mm apart.

Input Mating Connector 13a:

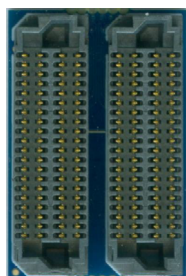
Female

Plexon Catalog Number:

CON/128f-Samtec Gen 1

Samtec Part Number:

ADF6-20-03.5-L-04-2 Gen 1



Input Connector 13a:

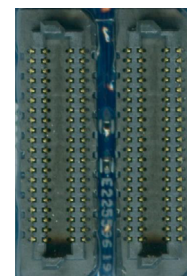
Male

Catalog Number:

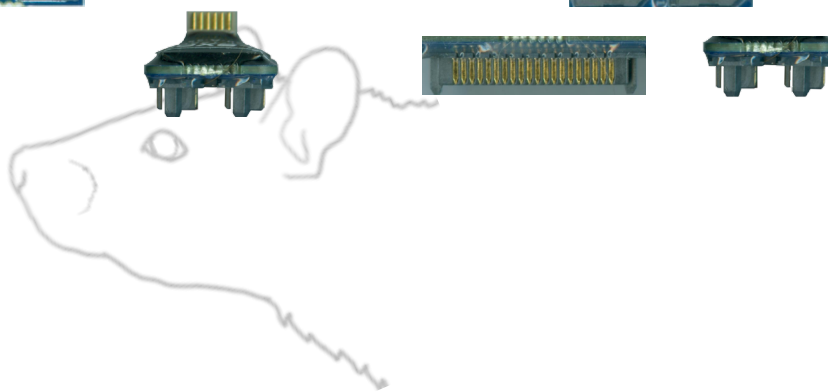
CON/128m-Samtec Gen 1

Samtec Part Number:

ADM6-20-01.5-L-04-2 Gen 1



Thumbtack Design:



Headstage Cable Information:

The HST/128DS Gen 1 headstage connects to the DHP unit through the HSC/DHSC2 or HSC/DHSC4 cable.

HST/128 Gen 1 is a legacy part. HST/128 Gen 2 headstages use a new Samtec input connectors. The main difference between the Gen 1 and Gen 2 input connectors is in the size and location of the guide post feature of the connectors.

In general, HST/128 Gen 2 headstages do not mate with accessories made with Gen 1 connectors and similarly Gen 1 headstages do not mate with accessories made with Gen 2 connectors.

Headstage V2- HST/128DS Gen 2 – 128 channel Digital Headstage (Samtec connector, thumbtack design)



Channels:	128	Amplifier/Processor Compatibility:	Digital Headstage Processor (DHP)
Connector:	Samtec (see below)	Dimensions:	0.83L x 0.47W x 0.39D inches (21L x 12W x 10D mm)
Reference pins:	16	Weight:	0.062oz (1.75g)
Grounding pins:	16	Power Supply:	Powered from Digital Headstage Processor (DHP)
Guide pins:	0	Input Impedance:	30M Ω @ 20kHz
Catalog Number:	HST/128DS Gen 2		
Reference:	Dedicated reference; programmably groundable		

Description:

A 128-channel thumbtack design digital headstage with a Samtec connector interface. HST/128DS Gen 2 is also alternatively available with on board red, green, or blue LEDs (in any combination) or with LED sockets for the ability to add LEDs in the future. Custom electrode interface boards are available for HST/128DS Gen 2.

Input Connector Information: Samtec connector #13b

Note: The input connector is a pair of Samtec ADM6-20-01.5-L-04-2 Gen 2 connectors. The connectors are spaced 6.35 mm apart.

Input Mating Connector 13b:

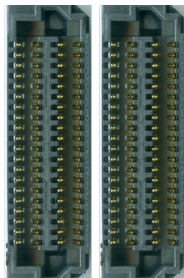
Female

Plexon Catalog Number:

CON/128f-Samtec Gen 2

Samtec Part Number:

ADF6-20-03.5-L-04-2 Gen 2



Input Connector 13b:

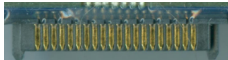
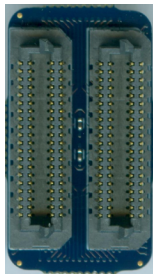
Male

Catalog Number:

CON/128m-Samtec Gen 2

Samtec Part Number:

ADM6-20-01.5-L-04-2 Gen 2



Thumbtack Design:

Headstage Cable Information:

The HST/128DS Gen 2 headstage connects to the DHP unit through the HSC/DHSC2 or HSC/DHSC4 cable.

HST/128 Gen 1 is a legacy part. HST/128 Gen 2 headstages use a new Samtec input connectors. The main difference between the Gen 1 and Gen 2 input connectors is in the size and location of the guide post feature of the connectors.

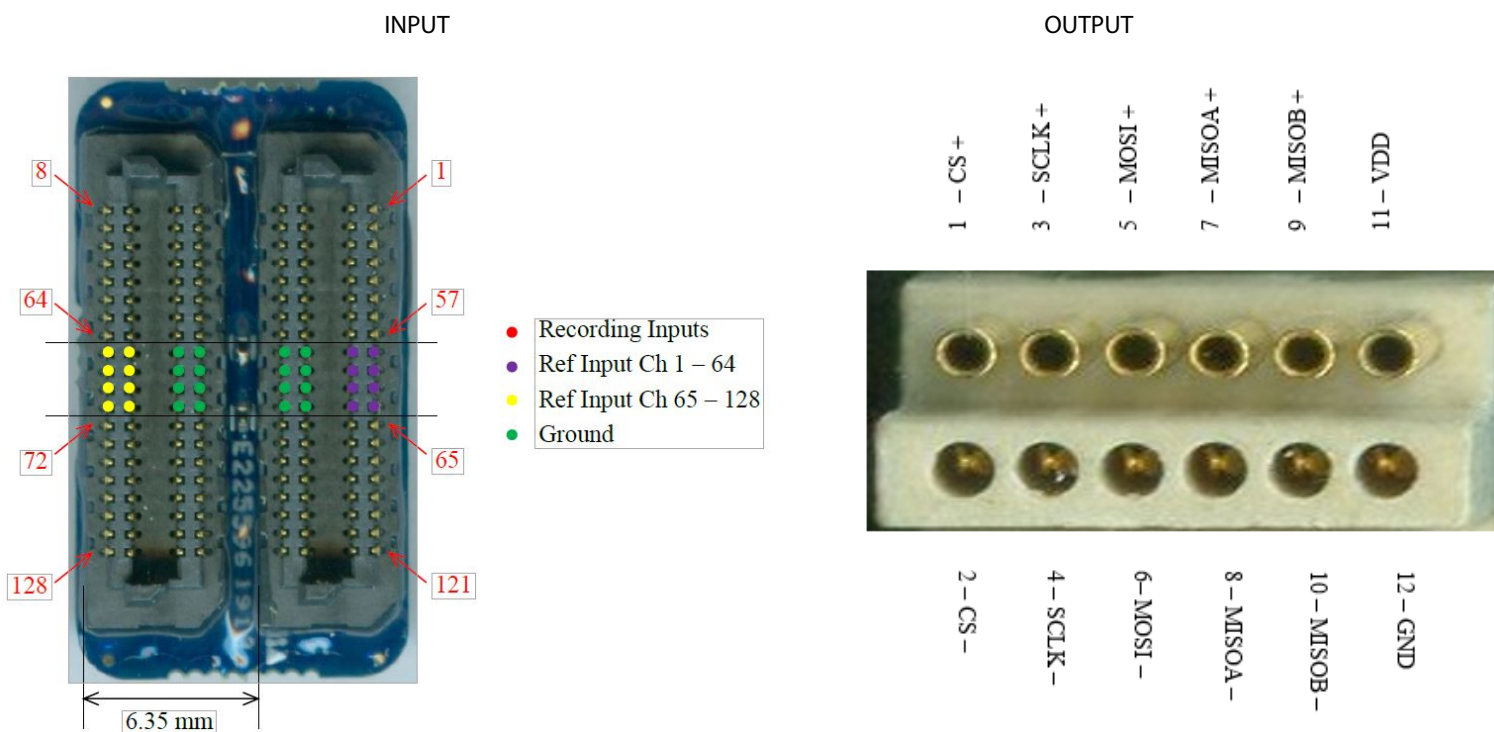
In general, HST/128 Gen 2 headstages do not mate with accessories made with Gen 1 connectors and similarly Gen 1 headstages do not mate with accessories made with Gen 2 connectors.

Connector Pin-out, Ground and Reference Information

Dedicated reference; programmably groundable

The figure on the left below shows which pins on the connector are recording inputs, which pins are connected to the (single) reference input, which pins are connected to ground, and which pins are not connected. The figure on the right below shows the Plexon channel numbers assigned to each of the recording inputs.

Catalog Number: HST/128DS (Gen 1 & 2)



Input Reference Channels 1-64:

- These inputs can be used to connect a dedicated reference electrode. They can be used as a point of reference for channels 1-64.
- Set the reference option in OmniPlex Server to "True Reference" to subtract from the reference electrode(s). It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- Set the reference option in OmniPlex Server to "Grounded Reference" to ground the input when no reference electrode(s) is present or when you want to disregard the signal from the reference electrode(s). This is typically used as a distant reference for field potentials, and also for spike channels.

Input Reference Channels 65-128:

- These inputs can be used to connect a dedicated reference electrode. They can be used as a point of reference for channels 65-128.
- Set the reference option in OmniPlex Server to "True Reference" to subtract from the reference electrode(s). It is typically connected to a de-insulated electrode and used as a local reference for spike channels.
- Set the reference option in OmniPlex Server to "Grounded Reference" to ground the input when no reference electrode(s) is present or when you want to disregard the signal from the reference electrode(s). This is typically used as a distant reference for field potentials, and also for spike channels.

Input Ground: Pins G1-16

- The ground on pins G1-G16 are identical. Use any or all of these pins to ground the animal via skull screw or cannula, and make sure it touches the cerebrospinal fluid.

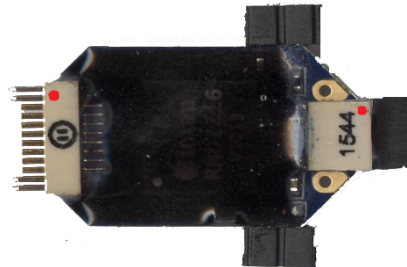
Note

- When using a version of OmniPlex prior to 17.0, please contact Plexon Support for a channel mapping file

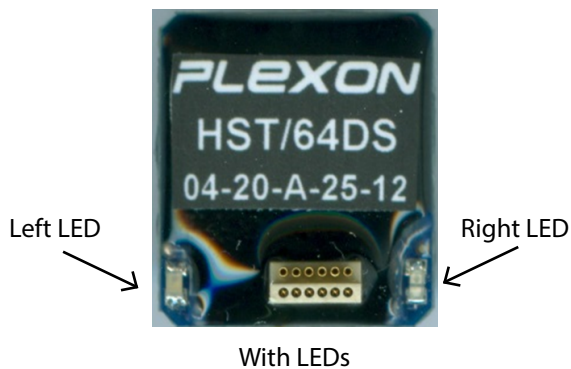
Headstage and Conversion Options

Light Emitting Diode (LED)

Plexon digital headstages can be equipped with mounted LEDs for use in behavioral analysis research using Plexon's CinePlex® Behavioral Research System with CinePlex Tracking. Red, blue, or green LEDs are available. The headstage pictured to the left is a 0.025" pitch 32 channel digital headstage with two LED sockets (HST/32D-2LED) and the headstage pictured on the right is a 0.025" pitch, 16 channel headstage with three LED sockets (HST/16D GEN2-3LED).

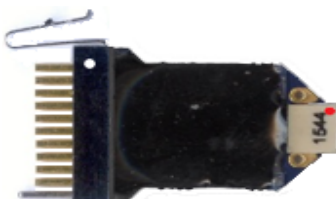


The HST/64DS can be ordered with LED sockets or with LEDs directly mounted to the headstage.



Latching

More active animals may require an especially secure connection between the mating connector and the headstage. Plexon's 0.050" pitch, digital headstages are available with a special clipping (latching) mechanism that delivers this additional security. The headstage pictured below is a 0.050" pitch, 8 digital channel headstage with the latching option (HST/8o50D Gen2-L).



DHP Connector Pinout, Ground and Reference Information

This section describes the connections and pinouts for the Digital Headstage Processor (DHP) unit. The front panel is shown in the image below.



Sig Com—Signal common is the local zero-voltage reference point of the DHP unit.

Earth—Earth is a direct connection to Earth ground. In the United States, Earth ground connects to the third prong on the wall outlet, which is eventually tied to hard earth at the power service entrance to the building.

NOTE: Ambient noise in buildings and noise radiated by electronic equipment are very common, and they can interfere with the signals in your experiment. To help reduce noise problems, you can connect the green ground wire from either Sig Com or Earth to metal object(s) near the animal being studied, such as headposts or apparatus framing or plates. It is best to try connecting to Sig Com and observing the noise reduction effect, then connecting instead to Earth and observing the effect, then comparing the results. Use the connection that gives the best results (best noise reduction).

Isolated/Grounded switch—The switch allows you to directly connect Sig Com to Earth ground or not. If Sig Com is not directly connected to Earth ground, the DHP unit is isolated from Earth ground.

HST PWR—The DHP unit is powered through the blue cable, which connects the DHP to the DATA LINK card in the OmniPlex D chassis. The DHP is powered on, and the PWR (power) light on the DHP unit is lit, when the OmniPlex Server application is running and a DHP topology is loaded.

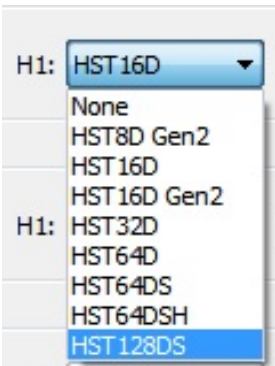
DHP Connector Pinout, Ground and Reference Information

The DHP unit can be supplied with up to four signal cards, each of which can handle up to 128 neural signal channels. The diagram below shows the pinouts for the individual DHP connectors. Connections are made to Plexon 8, 16, 32, and 64 channel digital headstages through the appropriate Plexon digital headstage cables.



Matching Digital Headstage Models to OmniPlex Server Settings

The dropdown list in the OmniPlex Server GUI allows you to identify the specific headstage attached to each connector on the DHP front panel:



When you are configuring the OmniPlex Server, use the following settings:

Headstage Model	Selection from Dropdown
HST/8o50D Gen2	HST8D Gen2
HST/16D	HST16D
HST/16o50D Gen2	HST16D Gen2
HST/16D Gen2	HST16D Gen2
HST/16D Gen2 DIFF	HST16D Gen2
HST/32D	HST32D
HST/64D	HST64D
HST/64DS	HST/64DS
HST/64DSH	HST/64DSH
HST/128DS (Gen 1 & 2)	HST/128DS

About Plexon Inc

Plexon is a pioneer and leading innovator of custom, high-performance data acquisition, behavior and analysis solutions specifically designed for scientific research. We collaborate with and supply thousands of customers including the most prestigious neuroscience laboratories around the globe driving new frontiers in areas including basic science, brain-machine interfaces (BMI), neurodegenerative diseases, addictive behaviors and neuroprosthetics. Plexon offers integrated solutions for in vivo neurophysiology, optogenetics, and behavioral research – backed by its industry-leading commitment to quality and customer support. For more information, please visit www.plexon.com.

Sales Support

For Sales Support, email info@plexon.com or call +1 (214) 369-4957.

Technical Support

If after reviewing this document, you would still like to access Plexon's Technical Support, we are available via several communication channels. You are invited to reach us through email or on the phone as follows:

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support@plexon.com

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8:30 a.m. to 5:00 p.m. Central Time
+1 (214) 369-4957